Near-Field ISAC for Six-Dimensional UAV Pose Estimation

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***Abstract*—This paper investigates near-field integrated sensing and communication (ISAC) for estimating the six-dimensional pose of a landing unmanned aerial vehicle (UAV). Conventional point-target sensing cannot recover the UAV orientation, while separately localizing multiple body scatterers ignores their rigid-body coupling. We consider an extremely large-scale array base station that communicates with the UAV while probing calibrated coded landmarks on its body. Orthogonal codes separate the landmark echoes, and the known landmark geometry couples all observations through a common position and orientation. Based on the exact spherical-wave response, we derive the pose equivalent Fisher information matrix after eliminating the unknown landmark reflectivities and establish a necessary and sufficient local identifiability condition. We then jointly optimize the communication and landmark-sensing powers under rate and total-power constraints, with the resulting weighted pose-bound minimization reformulated as an exact convex semidefinite program. Finally, a variable-projection estimator eliminates the reflectivities in closed form and recovers the absolute UAV pose. Numerical results validate the analysis and demonstrate the communication–pose-accuracy tradeoff.**

***Index Terms*—Integrated sensing and communication, near-field beam focusing, UAV landing, six-dimensional pose, Cramér-Rao bound, backscatter landmarks.**

I. Introduction

Integrated sensing and communication (ISAC) has been recognized as a key enabling technology for future wireless networks, by allowing radio-frequency hardware, spectrum, and transmitted waveforms to be reused for both communication and sensing [1], [2]. Besides improving resource utilization, ISAC can provide environmental and location awareness for various emerging applications. Autonomous landing of an unmanned aerial vehicle (UAV) is one such application, where a reliable control link and accurate knowledge of the UAV position and orientation are simultaneously required. With an extremely large-scale (XL) array deployed at the base station (BS), a low-altitude UAV may lie in the radiative near field, where spherical wavefronts enable angle-and-distance-dependent beam focusing [3], [4]. This makes near-field ISAC particularly promising for communication-assisted six-dimensional UAV pose estimation.

There have been extensive studies on ISAC transmit optimization. In [5], [6], the transmit beamforming was designed to balance the communication performance and sensing beampattern or Cramér–Rao bound (CRB) metrics, while the CRB–rate tradeoff for extended-target sensing was investigated in [7]. For near-field ISAC, prior works have mainly focused on point-target localization and beam-focusing design [4], [8]. Furthermore, UAV-enabled ISAC was studied in [9], where the UAV acts as an aerial platform and its maneuver is jointly designed with the transmit beamforming for communicating with users and sensing ground targets.

Despite the above works, however, the use of near-field ISAC for estimating the pose of a landing UAV has not been studied yet. Different from conventional UAV-enabled ISAC, the UAV considered here is simultaneously the communication terminal and a rigid sensing object. A point-target model cannot reveal its orientation, while independently localizing multiple body scatterers and then performing rigid registration fails to exploit their common pose and different echo qualities. In addition, the communication antenna and sensing landmarks are located at different positions on the UAV. Hence, a beam focused on the communication antenna does not generally illuminate all landmarks, whereas the landmark-focused signals share the transmit-power budget and interfere with communication reception. The unknown complex landmark reflectivities further act as nuisance parameters in pose estimation. These characteristics render existing point-target sensing and decoupled localization designs inapplicable to the considered problem.

Motivated by the above, this paper studies a monostatic near-field ISAC system, where an XL-array BS communicates with a landing UAV while probing coded backscatter landmarks installed at calibrated body-frame locations, as motivated by modulated retroreflective tags [10]. The array transmits one communication signal towards the UAV antenna and dedicated sensing signals towards the landmarks. The landmark codes separate and associate the reflected echoes, while the known rigid-body geometry couples all landmark coordinates through a common position and orientation. Under this setup, we aim to minimize a weighted CRB for the six-dimensional UAV pose by jointly optimizing the communication and landmark-sensing powers, subject to the communication-rate and total-power constraints. We further seek to recover the absolute UAV pose from the separated echoes in the presence of unknown landmark reflectivities. The main results are summarized as follows.

- First, we establish a near-field ISAC model for the coded rigid-body landmarks based on the exact spherical-wave response. We derive the pose-to-echo derivatives, construct the Fisher information matrix (FIM), and obtain the pose equivalent FIM (EFIM) and CRB by eliminating the unknown landmark reflectivities. We further characterize a necessary and sufficient condition for local pose identifiability, which reveals how the landmark geometry affects pose recovery.
- Next, we formulate the joint communication-and-sensing power allocation as a weighted pose-CRB minimization

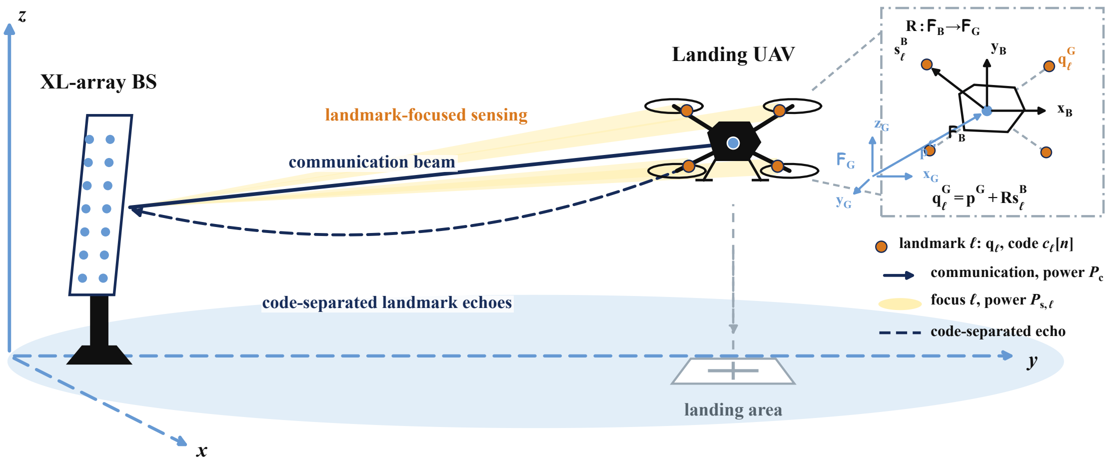


Fig. 1. Near-field ISAC system for six-dimensional UAV pose estimation.

problem, subject to the communication-rate and total-power constraints. Although the objective involves the inverse pose information matrix, we reformulate the problem as an exact convex semidefinite program, whose globally optimal solution can be efficiently obtained.

- Finally, we develop a variable-projection estimator for absolute UAV pose recovery, where the unknown landmark reflectivities are eliminated in closed form and the pose is iteratively refined. Numerical results validate the derived pose information and demonstrate the effectiveness of the proposed power allocation and pose estimator in balancing the communication–pose-accuracy tradeoff.

The remainder of this paper presents the system model, pose-information analysis, joint power allocation and pose estimation, numerical results, and conclusions.

Notation: Italic, bold lowercase, bold uppercase, and calligraphic letters denote scalars, vectors, matrices, and sets. The sets $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{S}_+^d$ denote real, complex, and positive-semidefinite domains. Superscripts T, H, and $*$ denote transpose, Hermitian transpose, and conjugate. The operators $\|\cdot\|_2$, $\mathrm{tr}(\cdot)$, $\mathrm{diag}(\cdot)$, $\Re\{\cdot\}$, $\Im\{\cdot\}$, $\mathrm{rank}(\cdot)$, and $\lambda_{\min}(\cdot)$ have their standard meanings. Moreover, $\mathbf{I}_d$, $\mathbf{1}_d$, $\mathbf{0}$, and δ_{ij} denote identity, all-ones, zero, and Kronecker-delta objects; $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a circular complex Gaussian distribution; and $\mathrm{SO}(3)$ and $\mathrm{SE}(3)$ denote the special orthogonal and special Euclidean groups in three dimensions, respectively.

II. System Model

In this section, we establish the near-field ISAC system model by specifying the rigid coded-landmark geometry, the shared communication-and-sensing transmission, and the code-separated monostatic echo observations.

A. Rigid Coded-Landmark Geometry

As illustrated in Fig. 1, the XL-array BS communicates with the landing UAV through a beam directed toward its antenna while focusing mutually orthogonal sensing probes on L calibrated landmarks rigidly attached to the UAV body. The landmark codes allow the BS to separate and associate the reflected echoes. The inset in Fig. 1 further illustrates how the UAV body frame is mapped to the global frame; this geometric relation is formalized next.

We attach a global frame $\mathcal{F}_{\mathrm{G}}$ to the XL-array BS and a body frame $\mathcal{F}_{\mathrm{B}}$ to the UAV, with the communication-antenna phase center chosen as the body-frame origin. Let $\mathbf{p} \in \mathbb{R}^3$ denote this origin in $\mathcal{F}_{\mathrm{G}}$, and let $\mathbf{R} \in \mathrm{SO}(3)$ map a vector expressed in $\mathcal{F}_{\mathrm{B}}$ to $\mathcal{F}_{\mathrm{G}}$. Since $\mathbf{R}^T\mathbf{R} = \mathbf{I}_3$ and $\det(\mathbf{R}) = 1$, its columns specify the three body-axis directions in the global frame. The physical UAV pose is therefore $\boldsymbol{\chi} = (\mathbf{p}, \mathbf{R})$.

The ℓth landmark has a calibrated body-frame coordinate $\mathbf{s}_\ell$ and a global coordinate $\mathbf{q}_\ell$. Since all landmarks are rigidly attached to the UAV, these coordinates satisfy [11]

$$\mathbf{q}_\ell = \mathbf{p} + \mathbf{R}\mathbf{s}_\ell, \qquad \ell \in \mathcal{L} \triangleq \{1, \ldots, L\}. \tag{1}$$

In (1), $\mathbf{p}$ translates every landmark by the same amount, whereas $\mathbf{R}\mathbf{s}_\ell$ rotates its known body-frame offset. Hence, all $\{\mathbf{q}_\ell\}$ are coupled through one common pose rather than being independent point locations. This rigid coupling is essential for recovering the UAV orientation from multiple landmark echoes.

Although $\mathbf{R}$ contains nine entries, its orthogonality and unit-determinant constraints leave only three rotational degrees of freedom. For estimation and CRB analysis, we represent this same rotation matrix by the z-y-x (ZYX) Euler-angle coordinates $\boldsymbol{\varphi} = [\phi, \theta, \psi]^T$, denoting roll, pitch, and yaw, respectively [12]. Let $c_\beta = \cos\beta$ and $s_\beta = \sin\beta$. Then

$$\mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{bmatrix}, \quad \mathbf{R}_y(\theta) = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix},$$
$$\mathbf{R}_z(\psi) = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R} = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi). \tag{2}$$

The multiplication order in (2) means that the roll rotation acts first, followed by the pitch and yaw rotations. The ZYX chart is locally nonsingular when $\cos\theta \neq 0$. We therefore collect the three position and three orientation coordinates into the six-dimensional absolute pose vector

$$\boldsymbol{\eta} \triangleq [\mathbf{p}^T, \boldsymbol{\varphi}^T]^T = [p_x, p_y, p_z, \phi, \theta, \psi]^T \in \mathbb{R}^6. \quad (3)$$

The vector $\boldsymbol{\eta}$, rather than the nine entries of $\mathbf{R}$, is the parameter estimated in the sequel. It contains $\mathbf{p}$ directly, determines $\mathbf{R}$ through (2), and gives each landmark coordinate as $\mathbf{q}_\ell = \mathbf{p} + \mathbf{R}\mathbf{s}_\ell$. Thus, estimating $\boldsymbol{\eta}$ recovers one common physical pose $(\mathbf{p}, \mathbf{R})$, rather than L unrelated landmark positions.

Each landmark also carries a known unit-modulus code $c_\ell[n]$. These codes label the landmarks and, together with the mutually orthogonal probing sequences introduced next, enable the BS to isolate the echo associated with each ℓ.

Finally, let $\mathbf{r}_m$ denote the global coordinate of BS array element m, where $m = 1, \ldots, M$, and let λ denote the carrier wavelength. The propagation distance from element m to a point $\mathbf{q}$ is $d_m(\mathbf{q}) = \|\mathbf{q} - \mathbf{r}_m\|$, and the wavenumber is $k_0 = 2\pi/\lambda$. Using element 1 as the phase reference and absorbing the common propagation loss and phase into the complex reflectivity, the normalized spherical-wave steering vector is

$$[\mathbf{a}(\mathbf{q})]_m = \frac{1}{\sqrt{M}} e^{-\jmath k_0 [d_m(\mathbf{q}) - d_1(\mathbf{q})]}. \quad (4)$$

B. ISAC Transmission and Communication Reception

The BS simultaneously transmits one data stream and L landmark-focused probing streams over the same time-frequency resource. Let $s[n]$ and $u_\ell[n]$ denote their normalized symbols, and let $\mathbf{v}_\mathrm{c}$ and $\mathbf{v}_\ell$ be unit-norm beams directed toward the UAV phase center and landmark ℓ, respectively. The transmit signal is

$$\begin{aligned} \mathbf{x}[n] = \sqrt{P_\mathrm{c}}\mathbf{v}_\mathrm{c} s[n] \\ + \sum_{\ell=1}^{L} \sqrt{P_{\mathrm{s},\ell}}\mathbf{v}_\ell u_\ell[n], \\ P_\mathrm{c} + \sum_{\ell=1}^{L} P_{\mathrm{s},\ell} \le P_{\max}. \end{aligned} \quad (5)$$

Let $\mathbf{h}_\mathrm{c}$ denote the downlink channel and $z_\mathrm{c}[n] \sim \mathcal{CN}(0, \sigma_\mathrm{c}^2)$ the receiver noise. Define the desired-link gain $\beta_\mathrm{c} \triangleq |\mathbf{h}_\mathrm{c}^H \mathbf{v}_\mathrm{c}|^2$ and the probing leakage gains $\beta_\ell \triangleq |\mathbf{h}_\mathrm{c}^H \mathbf{v}_\ell|^2$, with $\boldsymbol{\beta} \triangleq [\beta_1, \ldots, \beta_L]^T$. The UAV has no sensing-waveform cancellation stage and therefore treats the superposed probing streams as interference. Its received signal and achievable rate are

$$\begin{aligned} r_\mathrm{c}[n] = \sqrt{P_\mathrm{c}}\mathbf{h}_\mathrm{c}^H \mathbf{v}_\mathrm{c} s[n] \\ + \sum_{\ell=1}^{L} \sqrt{P_{\mathrm{s},\ell}}\mathbf{h}_\mathrm{c}^H \mathbf{v}_\ell u_\ell[n] \\ + z_\mathrm{c}[n], \\ R_\mathrm{c}(P_\mathrm{c}, \mathbf{P}_\mathrm{s}) = \log_2\left(1 + \frac{P_\mathrm{c}\beta_\mathrm{c}}{\sigma_\mathrm{c}^2 + \boldsymbol{\beta}^T \mathbf{P}_\mathrm{s}}\right), \end{aligned} \quad (6)$$

where $\mathbf{P}_\mathrm{s} \triangleq [P_{\mathrm{s},1}, \ldots, P_{\mathrm{s},L}]^T$. The waveform orthogonality introduced next enables matched filtering at the transmitting BS, but it does not remove the instantaneous probing interference at the UAV.

C. Code-Separated Monostatic Echo

The landmarks impose orthogonal signatures satisfying $N^{-1}\sum_{n=1}^{N} c_\ell[n] c_j^*[n] = \delta_{\ell j}$. The probing sequence $u_\ell[n]$ identifies the transmitted landmark-focused component, whereas $c_\ell[n]$ identifies the reflecting landmark. For compactness, define $\mathbf{a}_\ell \triangleq \mathbf{a}(\mathbf{q}_\ell)$, $g_\ell \triangleq \mathbf{a}_\ell^H \mathbf{v}_\ell$, and $\mathbf{b}_\ell \triangleq \mathbf{a}_\ell \mathbf{a}_\ell^H \mathbf{v}_\ell = g_\ell \mathbf{a}_\ell$. After direct-path and static-clutter cancellation, the array echo and the matched-filter output are

$$\mathbf{r}[n] = \sum_{j=1}^{L} \alpha_j c_j[n] \mathbf{a}_j \mathbf{a}_j^H \mathbf{x}[n] + \mathbf{z}[n], \quad (7)$$

$$\mathbf{y}_\ell \triangleq \frac{1}{N}\sum_{n=1}^{N} c_\ell^*[n] u_\ell^*[n] \mathbf{r}[n]. \quad (8)$$

where $\mathbf{z}[n]$ is the pre-processing receiver noise. Substituting (5) into $\mathbf{r}[n]$, correlation with $u_\ell[n]$ removes the communication term and all other probing components, while despreading with $c_\ell[n]$ removes returns from all other landmarks. The surviving term corresponds to (u_ℓ, c_ℓ), yielding

$$\mathbf{y}_\ell = \sqrt{P_{\mathrm{s},\ell}}\alpha_\ell \mathbf{b}_\ell + \mathbf{n}_\ell. \quad (9)$$

where $\alpha_\ell \in \mathbb{C}$ is the unknown reflectivity and $\mathbf{n}_\ell = N^{-1}\sum_{n=1}^{N} c_\ell^*[n] u_\ell^*[n] \mathbf{z}[n] \sim \mathcal{CN}(\mathbf{0}, \sigma_\mathrm{s}^2 \mathbf{I}_M)$ is the post-processing noise. Coding supplies association and inter-landmark separation, whereas orientation information arises from the nonzero offsets in (1).

III. CRB Analysis for Six-Dimensional Pose Estimation

This section derives the pose Fisher information from the separated landmark echoes, eliminates the unknown reflectivities, and establishes the geometric condition for locally identifiable six-dimensional pose estimation.

A. Real Parameter Vector and FIM

The parameter of interest is the absolute pose coordinate $\boldsymbol{\eta}$ defined in Section II-A; the nine entries of $\mathbf{R}$ are determined by its three Euler angles and are not estimated independently. For landmark ℓ, define the real nuisance vector $\boldsymbol{\rho}_\ell \triangleq [\Re\{\alpha_\ell\}, \Im\{\alpha_\ell\}]^T$. The complete real parameter vector is $\boldsymbol{\xi} \triangleq [\boldsymbol{\eta}^T, \boldsymbol{\rho}_1^T, \ldots, \boldsymbol{\rho}_L^T]^T \in \mathbb{R}^{6+2L}$.

From (9), the mean of $\mathbf{y}_\ell$ is

$$\boldsymbol{\mu}_\ell(\boldsymbol{\xi}) = \sqrt{P_{\mathrm{s},\ell}}\alpha_\ell \mathbf{b}_\ell. \quad (10)$$

The separated observations are independent and satisfy $\mathbf{y}_\ell \sim \mathcal{CN}(\boldsymbol{\mu}_\ell, \sigma_\mathrm{s}^2 \mathbf{I}_M)$. Hence, for $i, j \in \{1, \ldots, 6 + 2L\}$, the real FIM $\mathbf{J}_{\boldsymbol{\xi}}$ is given by

$$[\mathbf{J}_{\boldsymbol{\xi}}]_{i,j} = \frac{2}{\sigma_\mathrm{s}^2}\Re\left\{\sum_{\ell=1}^{L}\left(\frac{\partial \boldsymbol{\mu}_\ell}{\partial \xi_i}\right)^H \frac{\partial \boldsymbol{\mu}_\ell}{\partial \xi_j}\right\}. \quad (11)$$

It remains to derive the nonzero mean Jacobians in (11) and remove the reflectivity nuisance parameters.

B. Closed-Form Pose CRB

We proceed in six steps, from the required mean Jacobians and their pose and echo factors to FIM assembly, nuisance elimination, and the pose error bound. All derivatives are evaluated at the operating pose.

1) Required mean Jacobians: The pose dependence follows the chain $\boldsymbol{\eta} \to \mathbf{q}_\ell \to \mathbf{b}_\ell \to \boldsymbol{\mu}_\ell$. Define $\mathbf{H}_{\eta,\ell} \triangleq \partial \mathbf{q}_\ell / \partial \boldsymbol{\eta}^T \in \mathbb{R}^{3\times 6}$ and $\mathbf{G}_\ell \triangleq \partial \mathbf{b}_\ell / \partial \mathbf{q}_\ell^T \in \mathbb{C}^{M\times 3}$. Since $d\alpha_\ell = [1, \jmath] d\boldsymbol{\rho}_\ell$, the total differential of (10) yields

$$\begin{aligned} \frac{\partial \boldsymbol{\mu}_\ell}{\partial \boldsymbol{\eta}^T} &= \sqrt{P_{\mathrm{s},\ell}}\,\alpha_\ell \mathbf{G}_\ell \mathbf{H}_{\eta,\ell}, \\ \frac{\partial \boldsymbol{\mu}_\ell}{\partial \boldsymbol{\rho}_\ell^T} &= \sqrt{P_{\mathrm{s},\ell}}[\,\mathbf{b}_\ell, \jmath \mathbf{b}_\ell\,]. \end{aligned} \tag{12}$$

For $j \neq \ell$, $\partial \boldsymbol{\mu}_\ell / \partial \boldsymbol{\rho}_j^T = \mathbf{0}$. We next derive $\mathbf{H}_{\eta,\ell}$ and $\mathbf{G}_\ell$.

2) Pose-to-landmark Jacobian: Let $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ denote the Cartesian basis vectors. Differentiating the ZYX rotation in (2) and writing each rotation-induced landmark displacement as an ordinary cross product gives

$$\begin{aligned} \mathbf{B}(\boldsymbol{\varphi}) &\triangleq \left[\,\mathbf{e}_x \;\; \mathbf{R}_x^T(\phi)\mathbf{e}_y \;\; \mathbf{R}_x^T(\phi)\mathbf{R}_y^T(\theta)\mathbf{e}_z\,\right], \\ \mathbf{C}_\ell &\triangleq \left[\,\mathbf{e}_x \times \mathbf{s}_\ell \;\; \mathbf{e}_y \times \mathbf{s}_\ell \;\; \mathbf{e}_z \times \mathbf{s}_\ell\,\right], \\ d\mathbf{q}_\ell &= d\mathbf{p} + \mathbf{R}\big(\mathbf{C}_\ell\big(\mathbf{B}(\boldsymbol{\varphi})d\boldsymbol{\varphi}\big)\big) = \mathbf{H}_{\eta,\ell} d\boldsymbol{\eta}, \\ \mathbf{H}_{\eta,\ell} &= \left[\,\mathbf{I}_3 \;\; \mathbf{R}(\mathbf{C}_\ell \mathbf{B}(\boldsymbol{\varphi}))\,\right]. \end{aligned} \tag{13}$$

The columns of $\mathbf{B}(\boldsymbol{\varphi})$ are the body-frame roll, pitch, and yaw axes, while those of $\mathbf{C}_\ell$ are the corresponding unit-axis lever-arm displacements. Thus, the rotational block is evaluated from right to left and vanishes when $\mathbf{s}_\ell = \mathbf{0}$.

3) Landmark-to-echo Jacobian: For $m \in \{1, \ldots, M\}$, define the unit vector from array element m to landmark ℓ as $\mathbf{u}_{\ell m} \triangleq (\mathbf{q}_\ell - \mathbf{r}_m)/d_m(\mathbf{q}_\ell)$. Differentiating (4) gives

$$\mathbf{D}_\ell \triangleq \frac{\partial \mathbf{a}_\ell}{\partial \mathbf{q}_\ell^T}, \qquad [\mathbf{D}_\ell]_{m,:} = -\jmath k_0 [\mathbf{a}_\ell]_m (\mathbf{u}_{\ell m} - \mathbf{u}_{\ell 1})^T, \tag{14}$$

where $\mathbf{D}_\ell \in \mathbb{C}^{M\times 3}$ and $d\mathbf{a}_\ell = \mathbf{D}_\ell d\mathbf{q}_\ell$. The focal vector $\mathbf{v}_\ell$ is fixed during differentiation. Therefore, $dg_\ell = (\mathbf{D}_\ell^H \mathbf{v}_\ell)^T d\mathbf{q}_\ell$. Using $\mathbf{b}_\ell = g_\ell \mathbf{a}_\ell$ gives $d\mathbf{b}_\ell = \mathbf{G}_\ell d\mathbf{q}_\ell$, where

$$\mathbf{G}_\ell = g_\ell \mathbf{D}_\ell + \mathbf{a}_\ell (\mathbf{D}_\ell^H \mathbf{v}_\ell)^T. \tag{15}$$

Equations (13) and (15) provide the two factors required in the first line of (12).

4) FIM assembly: Substitution of (12) into (11) yields the three nonzero blocks contributed by landmark ℓ:

$$\begin{aligned} \mathbf{J}_{\eta\eta,\ell} &= \frac{2P_{\mathrm{s},\ell}|\alpha_\ell|^2}{\sigma_{\mathrm{s}}^2} \mathbf{H}_{\eta,\ell}^T \Re\{\mathbf{G}_\ell^H \mathbf{G}_\ell\} \mathbf{H}_{\eta,\ell}, \\ \mathbf{J}_{\eta\rho,\ell} &= \frac{2P_{\mathrm{s},\ell}}{\sigma_{\mathrm{s}}^2} \Re\left\{\alpha_\ell^* \mathbf{H}_{\eta,\ell}^T \mathbf{G}_\ell^H [\,\mathbf{b}_\ell, \jmath \mathbf{b}_\ell\,]\right\}, \\ \mathbf{J}_{\rho\rho,\ell} &= \frac{2P_{\mathrm{s},\ell}\|\mathbf{b}_\ell\|_2^2}{\sigma_{\mathrm{s}}^2} \mathbf{I}_2. \end{aligned} \tag{16}$$

These blocks have dimensions 6×6, 6×2, and 2×2, respectively; all cross-landmark nuisance blocks are zero.

5) Reflectivity elimination: The nuisance FIM is block diagonal across landmarks. Taking its Schur complement gives the Euler-coordinate pose EFIM

$$\mathbf{J}_\eta \triangleq \sum_{\ell=1}^{L} \left(\mathbf{J}_{\eta\eta,\ell} - \mathbf{J}_{\eta\rho,\ell} \mathbf{J}_{\rho\rho,\ell}^{-1} \mathbf{J}_{\eta\rho,\ell}^T \right). \tag{17}$$

Evaluating the Schur complement in closed form gives

$$\begin{aligned} \boldsymbol{\Pi}_{\mathbf{b}_\ell}^\perp &\triangleq \mathbf{I}_M - \frac{\mathbf{b}_\ell \mathbf{b}_\ell^H}{\|\mathbf{b}_\ell\|_2^2}, \\ \bar{\mathbf{K}}_\ell &\triangleq \frac{2|\alpha_\ell|^2}{\sigma_{\mathrm{s}}^2} \Re\left\{\mathbf{G}_\ell^H \boldsymbol{\Pi}_{\mathbf{b}_\ell}^\perp \mathbf{G}_\ell\right\}, \\ \mathbf{J}_\eta &= \sum_{\ell=1}^{L} P_{\mathrm{s},\ell} \mathbf{H}_{\eta,\ell}^T \bar{\mathbf{K}}_\ell \mathbf{H}_{\eta,\ell}. \end{aligned} \tag{18}$$

Here, $\boldsymbol{\Pi}_{\mathbf{b}_\ell}^\perp$ removes the echo variation parallel to $\mathbf{b}_\ell$, which is indistinguishable from a reflectivity change. The real symmetric matrix $\bar{\mathbf{K}}_\ell \succeq \mathbf{0}$ is the nuisance-free Cartesian information per unit sensing power for landmark ℓ.

6) Pose CRB and scalar metric: Let $\widehat{\boldsymbol{\eta}}$ denote a locally unbiased estimate of the absolute pose coordinates and define $\Delta\boldsymbol{\eta} \triangleq \widehat{\boldsymbol{\eta}} - \boldsymbol{\eta}$ only as its estimation error. The CRB is $\mathbb{E}\{\Delta\boldsymbol{\eta}\Delta\boldsymbol{\eta}^T\} \succeq \mathbf{J}_\eta^{-1}$. Accordingly, the position error bound (PEB) and orientation error bound (OEB) are

$$\begin{aligned} \mathrm{PEB} &= \sqrt{\mathrm{tr}\big([\mathbf{J}_\eta^{-1}]_{1:3,1:3}\big)}, \\ \mathrm{OEB} &= \sqrt{\mathrm{tr}\big([\mathbf{J}_\eta^{-1}]_{4:6,4:6}\big)}. \end{aligned} \tag{19}$$

The OEB measures the joint local estimation error of the ZYX roll, pitch, and yaw angles in radians. We consider regular landing attitudes with $\cos\theta \neq 0$, for which the ZYX chart and $\mathbf{B}(\boldsymbol{\varphi})$ in (13) are nonsingular. Let $\mathbf{P}_{\mathrm{s}} \triangleq [P_{\mathrm{s},1}, \ldots, P_{\mathrm{s},L}]^T$ and choose $w_{\mathrm{p}}, w_{\mathrm{o}} > 0$. The scalar pose-accuracy metric is

$$\begin{aligned} \mathbf{W} &\triangleq \mathrm{diag}(w_{\mathrm{p}}\mathbf{I}_3, w_{\mathrm{o}}\mathbf{I}_3) \succ \mathbf{0}, \\ \Phi(\mathbf{P}_{\mathrm{s}}) &\triangleq \mathrm{tr}\big(\mathbf{W}\mathbf{J}_\eta^{-1}(\mathbf{P}_{\mathrm{s}})\big). \end{aligned} \tag{20}$$

C. Local Pose Identifiability

Define the stacked pose-sensitivity matrix over all L landmarks as

$$\mathbf{F} \triangleq \begin{bmatrix} \sqrt{P_{\mathrm{s},1}}\,\bar{\mathbf{K}}_1^{1/2}\mathbf{H}_{\eta,1} \\ \vdots \\ \sqrt{P_{\mathrm{s},L}}\,\bar{\mathbf{K}}_L^{1/2}\mathbf{H}_{\eta,L} \end{bmatrix} \in \mathbb{R}^{3L\times 6}, \tag{21}$$

where $\bar{\mathbf{K}}_\ell^{1/2}$ is the unique positive-semidefinite square root of $\bar{\mathbf{K}}_\ell$.

Proposition 1: The pose CRB is finite if and only if

$$\mathrm{rank}(\mathbf{F}) = 6. \tag{22}$$

If, in addition, $\cos\theta \neq 0$ and $P_{\mathrm{s},\ell} > 0$ and $\bar{\mathbf{K}}_\ell \succ \mathbf{0}$ for every $\ell \in \mathcal{L}$, then (22) is equivalent to requiring at least three noncollinear landmark offsets.

Proof: By (18) and (21),

$$\mathbf{J}_\eta = \mathbf{F}^T\mathbf{F}. \tag{23}$$

Since $\mathrm{rank}(\mathbf{F}^T\mathbf{F}) = \mathrm{rank}(\mathbf{F})$, the 6×6 matrix $\mathbf{J}_\eta$ is nonsingular if and only if (22) holds. This proves the first statement.

We next prove the geometric statement. Under the additional conditions, $\sqrt{P_{\mathrm{s},\ell}}\bar{\mathbf{K}}_\ell^{1/2}$ is nonsingular for every $\ell \in \mathcal{L}$. Therefore, $\mathbf{F}$ has full column rank if and only if the only vector satisfying $\mathbf{H}_{\eta,\ell}\boldsymbol{\zeta} = \mathbf{0}$ for all $\ell \in \mathcal{L}$ is $\boldsymbol{\zeta} = \mathbf{0}$. Write

$\boldsymbol{\zeta} = [\boldsymbol{\zeta}_{\mathrm{p}}^T, \boldsymbol{\zeta}_{\varphi}^T]^T$ and define $\boldsymbol{\omega} \triangleq \mathbf{B}(\boldsymbol{\varphi})\boldsymbol{\zeta}_{\varphi}$. Since $\cos\theta \neq 0$, $\mathbf{B}(\boldsymbol{\varphi})$ is nonsingular. Equation (13) then gives

$$\mathbf{H}_{\eta,\ell}\boldsymbol{\zeta} = \boldsymbol{\zeta}_{\mathrm{p}} + \mathbf{R}(\boldsymbol{\omega} \times \mathbf{s}_\ell). \tag{24}$$

Choose three noncollinear offsets $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3$. If (24) is zero for all three, subtracting its first instance from the other two yields $\boldsymbol{\omega} \times (\mathbf{s}_2 - \mathbf{s}_1) = \mathbf{0}$ and $\boldsymbol{\omega} \times (\mathbf{s}_3 - \mathbf{s}_1) = \mathbf{0}$. The two offset differences are linearly independent, so $\boldsymbol{\omega} = \mathbf{0}$. Consequently, $\boldsymbol{\zeta}_{\varphi} = \mathbf{0}$ and $\boldsymbol{\zeta}_{\mathrm{p}} = \mathbf{0}$, proving sufficiency.

Conversely, if no three offsets are noncollinear, all of them lie on one line. Choose a nonzero $\boldsymbol{\omega}$ parallel to that line and set $\boldsymbol{\zeta}_{\varphi} = \mathbf{B}^{-1}(\boldsymbol{\varphi})\boldsymbol{\omega}$ and $\boldsymbol{\zeta}_{\mathrm{p}} = -\mathbf{R}(\boldsymbol{\omega} \times \mathbf{s}_1)$. Then $\mathbf{H}_{\eta,\ell}\boldsymbol{\zeta} = \mathbf{R}\big(\boldsymbol{\omega} \times (\mathbf{s}_\ell - \mathbf{s}_1)\big) = \mathbf{0}$ for every landmark, which gives a nonzero null vector of $\mathbf{F}$. For coincident offsets, any nonzero $\boldsymbol{\omega}$ works. This proves necessity and completes the proof. $\square$

Remark 1: The rank condition in Proposition 1 remains valid when some sensing powers are zero or some $\bar{\mathbf{K}}_\ell$ are rank deficient, because their effects are already included in $\mathbf{F}$. Under positive sensing powers and full Cartesian information, three landmarks are the minimum, but their offsets must form a nondegenerate triangle. Nearly collinear offsets or short rotational lever arms can still make $\mathbf{J}_\eta$ poorly conditioned; hence, both sensing strength and landmark geometry should guide the power allocation.

IV. Joint ISAC Power Allocation and Pose Estimation

This section first obtains the communication and sensing power allocation that minimizes the pose-estimation bound under the communication-rate and total-power requirements. It then estimates the absolute UAV pose from the separated landmark echoes.

A. Joint Communication-and-Sensing Power Allocation

The communication beam $\mathbf{v}_{\mathrm{c}}$ and landmark-probing beams $\{\mathbf{v}_\ell\}$ are fixed as specified in Section II. Hence, only the communication power P_{c} and sensing-power vector $\mathbf{P}_{\mathrm{s}}$ remain to be optimized.

1) Original problem: Let $R_{\min}$ denote the required communication rate. Moreover, $\underline{P}_\ell > 0$ is the tracking-power floor that preserves the ℓth coded landmark, while $\bar{P}_\ell$ is its maximum sensing power. Based on the weighted pose-CRB metric in (20), the original joint power-allocation problem is

$$\begin{aligned} \underset{P_{\mathrm{c}}, \mathbf{P}_{\mathrm{s}}}{\text{minimize}} \quad & \Phi(\mathbf{P}_{\mathrm{s}}) && \text{(25a)}\\ \text{subject to} \quad & R_{\mathrm{c}}(P_{\mathrm{c}}, \mathbf{P}_{\mathrm{s}}) \geq R_{\min}, && \text{(25b)}\\ & P_{\mathrm{c}} + \mathbf{1}_L^T \mathbf{P}_{\mathrm{s}} \leq P_{\max}, && \text{(25c)}\\ & P_{\mathrm{c}} \geq 0, \quad \underline{P}_\ell \leq P_{\mathrm{s},\ell} \leq \bar{P}_\ell, \quad \ell \in \mathcal{L}. && \text{(25d)} \end{aligned}$$

Problem (25) directly captures the communication–sensing tradeoff: increasing a sensing power improves the pose information, but consumes the common power budget and creates additional interference at the UAV receiver.

2) Elimination of the communication power: For any fixed $\mathbf{P}_{\mathrm{s}}$, the objective in (25a) is independent of P_{c}, whereas the rate in (6) is monotonically increasing in P_{c}. Therefore, an optimal solution can always use the minimum communication power that satisfies the rate constraint. Defining $\gamma_{\min} \triangleq 2^{R_{\min}} - 1$, this power is

$$P_{\mathrm{c}}^{\min}(\mathbf{P}_{\mathrm{s}}) = \frac{\gamma_{\min}}{\beta_{\mathrm{c}}} \left(\sigma_{\mathrm{c}}^2 + \boldsymbol{\beta}^T \mathbf{P}_{\mathrm{s}} \right). \tag{26}$$

Substituting (26) into the total-power constraint yields the equivalent sensing-power budget

$$\left(\mathbf{1}_L + \frac{\gamma_{\min}}{\beta_{\mathrm{c}}} \boldsymbol{\beta} \right)^T \mathbf{P}_{\mathrm{s}} \leq P_{\max} - \frac{\gamma_{\min}\sigma_{\mathrm{c}}^2}{\beta_{\mathrm{c}}}. \tag{27}$$

The coefficient $1 + \gamma_{\min}\beta_\ell/\beta_{\mathrm{c}}$ is the effective cost of allocating one unit of power to landmark ℓ: in addition to its direct power consumption, the resulting probing leakage requires extra communication power to maintain the target rate.

3) SDP reformulation and solution: The remaining difficulty is the inverse information matrix in $\Phi(\mathbf{P}_{\mathrm{s}})$. For any identifiable allocation satisfying $\mathbf{J}_\eta(\mathbf{P}_{\mathrm{s}}) \succ \mathbf{0}$, introduce $\mathbf{Z} \in \mathbb{S}_+^6$. By the Schur complement,

$$\mathbf{Z} \succeq \mathbf{J}_\eta^{-1}(\mathbf{P}_{\mathrm{s}}) \iff \begin{bmatrix} \mathbf{J}_\eta(\mathbf{P}_{\mathrm{s}}) & \mathbf{I}_6 \\ \mathbf{I}_6 & \mathbf{Z} \end{bmatrix} \succeq \mathbf{0}. \tag{28}$$

Since $\mathbf{W} \succ \mathbf{0}$, minimizing $\mathrm{tr}(\mathbf{WZ})$ makes the above inequality tight. Consequently, (25) is exactly equivalent to

$$\begin{aligned} \underset{\mathbf{P}_{\mathrm{s}}, \mathbf{Z} \in \mathbb{S}_+^6}{\text{minimize}} \quad & \mathrm{tr}(\mathbf{WZ}) && \text{(29a)}\\ \text{subject to} \quad & \begin{bmatrix} \mathbf{J}_\eta(\mathbf{P}_{\mathrm{s}}) & \mathbf{I}_6 \\ \mathbf{I}_6 & \mathbf{Z} \end{bmatrix} \succeq \mathbf{0}, && \text{(29b)}\\ & \left(\mathbf{1}_L + \frac{\gamma_{\min}}{\beta_{\mathrm{c}}} \boldsymbol{\beta} \right)^T \mathbf{P}_{\mathrm{s}} \leq P_{\max} - \frac{\gamma_{\min}\sigma_{\mathrm{c}}^2}{\beta_{\mathrm{c}}}, && \text{(29c)}\\ & \underline{P}_\ell \leq P_{\mathrm{s},\ell} \leq \bar{P}_\ell, \quad \ell \in \mathcal{L}. && \text{(29d)} \end{aligned}$$

By (18), $\mathbf{J}_\eta(\mathbf{P}_{\mathrm{s}})$ is affine in $\mathbf{P}_{\mathrm{s}}$. Therefore, (29) is a convex semidefinite program and can be solved globally using a standard interior-point solver. Denoting its solution by $(\mathbf{P}_{\mathrm{s}}^\star, \mathbf{Z}^\star)$, the optimal communication power and epigraph matrix are recovered as

$$\begin{aligned} P_{\mathrm{c}}^\star &= P_{\mathrm{c}}^{\min}(\mathbf{P}_{\mathrm{s}}^\star), \\ \mathbf{Z}^\star &= \mathbf{J}_\eta^{-1}(\mathbf{P}_{\mathrm{s}}^\star). \end{aligned} \tag{30}$$

Thus, the two transformations are exact, and $(P_{\mathrm{c}}^\star, \mathbf{P}_{\mathrm{s}}^\star)$ globally solves the original problem. The linear matrix inequality in (29b) also enforces the identifiability condition in Proposition 1; it becomes infeasible if the available landmark geometry and powers cannot yield a nonsingular pose FIM.

B. Absolute Pose Estimation Algorithm

For a trial absolute pose $\boldsymbol{\eta} = [\mathbf{p}^T, \boldsymbol{\varphi}^T]^T$, construct $\mathbf{R}(\boldsymbol{\varphi})$ using (2), $\mathbf{q}_\ell(\boldsymbol{\eta}) = \mathbf{p} + \mathbf{R}(\boldsymbol{\varphi})\mathbf{s}_\ell$, and the corresponding echo signature $\mathbf{b}_\ell(\boldsymbol{\eta})$. For this trial pose, the least-squares reflectivity estimate and the associated orthogonal projector are

$$\begin{aligned} \widehat{\alpha}_\ell(\boldsymbol{\eta}) &\triangleq \frac{\mathbf{b}_\ell^H(\boldsymbol{\eta})\mathbf{y}_\ell}{\sqrt{P_{\mathrm{s},\ell}^\star}\, \|\mathbf{b}_\ell(\boldsymbol{\eta})\|_2^2}, \\ \boldsymbol{\Pi}_\ell^\perp(\boldsymbol{\eta}) &\triangleq \mathbf{I}_M - \frac{\mathbf{b}_\ell(\boldsymbol{\eta})\mathbf{b}_\ell^H(\boldsymbol{\eta})}{\|\mathbf{b}_\ell(\boldsymbol{\eta})\|_2^2}. \end{aligned} \tag{31}$$

Algorithm 1 Variable-Projection Absolute Pose Estimation

Require: Separated echoes $\{\mathbf{y}_\ell\}$, optimized powers $\{P^{\star}_{\mathrm{s},\ell}\}$, calibrated offsets $\{\mathbf{s}_\ell\}$, initial absolute pose $\boldsymbol{\eta}^{(0)}$, and tolerance ϵ.

```
Set k = 0.
repeat
    for ℓ = 1, ..., L do
        Compute q_ℓ^(k), b_ℓ^(k), G_ℓ^(k), H_{η,ℓ}^(k), and Q_ℓ^(k).
        Update α̂_ℓ^(k) and form e_ℓ^(k) and T_ℓ^(k).
    end for
    Form Ĵ^(k) and d^(k) from (34).
    Solve (35), select ρ_k by backtracking, and update (36).
    Set k ← k + 1.
until ||η^(k) − η^(k−1)||_2 ≤ ϵ
Recover p̂, R̂, and q̂_ℓ = p̂ + R̂ s_ℓ.
```

Ensure: Absolute UAV pose, landmark coordinates, and reflectivities.

Eliminating the unknown reflectivities gives the concentrated maximum-likelihood estimator

$$\widehat{\boldsymbol{\eta}} = \underset{\boldsymbol{\eta}\in\mathbb{R}^6}{\arg\min}\ F(\boldsymbol{\eta}) \triangleq \sum_{\ell=1}^{L} \frac{\|\boldsymbol{\Pi}_\ell^{\perp}(\boldsymbol{\eta})\mathbf{y}_\ell\|_2^2}{\sigma_{\mathrm{s}}^2}, \tag{32}$$

where the search is restricted to the nonsingular ZYX chart specified in Section II-A.

We solve (32) by a variable-projection Gauss–Newton method. At iteration k, evaluate $\mathbf{b}_\ell^{(k)}$, $\mathbf{G}_\ell^{(k)}$, and $\mathbf{H}_{\eta,\ell}^{(k)}$ at $\boldsymbol{\eta}^{(k)}$, and define

$$\begin{aligned}
\mathbf{Q}_\ell^{(k)} &\triangleq \mathbf{G}_\ell^{(k)}\mathbf{H}_{\eta,\ell}^{(k)},\\
\mathbf{e}_\ell^{(k)} &\triangleq \mathbf{y}_\ell - \sqrt{P^{\star}_{\mathrm{s},\ell}}\,\widehat{\alpha}_\ell^{(k)}\mathbf{b}_\ell^{(k)} = \boldsymbol{\Pi}_\ell^{\perp(k)}\mathbf{y}_\ell,\\
\mathbf{T}_\ell^{(k)} &\triangleq \sqrt{P^{\star}_{\mathrm{s},\ell}}\,\widehat{\alpha}_\ell^{(k)}\boldsymbol{\Pi}_\ell^{\perp(k)}\mathbf{Q}_\ell^{(k)}\\
&\quad + \frac{\mathbf{b}_\ell^{(k)}\left((\mathbf{Q}_\ell^{(k)})^H\mathbf{e}_\ell^{(k)}\right)^T}{\|\mathbf{b}_\ell^{(k)}\|_2^2}.
\end{aligned} \tag{33}$$

With the sign convention in (33), $\mathbf{T}_\ell^{(k)}$ is the exact first-order Jacobian of the concentrated residual. Its first term projects the pose-to-echo Jacobian onto the complement of the unknown reflectivity, while its second term accounts for the change of the least-squares reflectivity with the trial pose. The latter is proportional to the residual and vanishes in the noise-free limit. The local linearization is $\mathbf{e}_\ell(\boldsymbol{\eta}^{(k)}+\Delta\boldsymbol{\eta}) \simeq \mathbf{e}_\ell^{(k)} - \mathbf{T}_\ell^{(k)}\Delta\boldsymbol{\eta}$. Accordingly, define

$$\begin{aligned}
\widehat{\mathbf{J}}^{(k)} &\triangleq \frac{2}{\sigma_{\mathrm{s}}^2}\sum_{\ell=1}^{L}\Re\Big\{(\mathbf{T}_\ell^{(k)})^H\mathbf{T}_\ell^{(k)}\Big\},\\
\mathbf{d}^{(k)} &\triangleq \frac{2}{\sigma_{\mathrm{s}}^2}\sum_{\ell=1}^{L}\Re\Big\{(\mathbf{T}_\ell^{(k)})^H\mathbf{e}_\ell^{(k)}\Big\}.
\end{aligned} \tag{34}$$

A damped Gauss–Newton direction is obtained from

$$\left(\widehat{\mathbf{J}}^{(k)} + \kappa_k\mathbf{I}_6\right)\Delta\boldsymbol{\eta}^{(k)} = \mathbf{d}^{(k)}, \qquad \kappa_k \geq 0, \tag{35}$$

and the absolute pose coordinates are updated as

$$\boldsymbol{\eta}^{(k+1)} = \boldsymbol{\eta}^{(k)} + \rho_k\Delta\boldsymbol{\eta}^{(k)}, \qquad 0 < \rho_k \leq 1, \tag{36}$$

where ρ_k is selected by backtracking to decrease $F(\boldsymbol{\eta})$. The rotation matrix is reconstructed from the updated Euler angles by (2); hence, it satisfies $\mathbf{R}^T\mathbf{R} = \mathbf{I}_3$ at every iteration. The increment $\Delta\boldsymbol{\eta}^{(k)}$ is only the numerical search direction, whereas the final estimate is the absolute pose $\widehat{\boldsymbol{\eta}}$.

As the concentrated residual decreases, the second term of $\mathbf{T}_\ell^{(k)}$ vanishes and $\widehat{\mathbf{J}}^{(k)}$ approaches the pose EFIM evaluated at the estimated reflectivities. Thus, the power-allocation design and the estimator use the same local information geometry: the former allocates power according to the predicted pose information, while the latter uses the measured echoes to approach it.

V. Numerical Results

This section presents numerical results to evaluate the proposed power allocation and pose estimator and to validate the preceding analytical results.

A. Simulation Setup and Benchmark Schemes

We consider a monostatic near-field ISAC system operating at 28 GHz. The BS employs a 256×256 half-wavelength uniform planar array (UPA), whose aperture is approximately 1.37 m on each side. One communication beam is directed toward the UAV phase center, while four landmark-focused sensing beams transmit mutually orthogonal length-four codes over 256 coherent snapshots. The UAV phase center is located at $[2.00, -1.00, 20.00]^T$ m. Its calibrated body-frame landmarks are $[0.81, 0.66, 0]^T$, $[-0.81, 0.63, 0]^T$, $[-0.69, -0.60, -0.225]^T$, and $[0.75, -0.57, 0.135]^T$ m, corresponding to a body radius of 1.04 m.

The normalized average echo powers of the four landmarks are $[1.00, 0.60, 0.10, 0.001]$. The resulting 30-dB visibility difference models the asymmetric self-shadowing that can occur during a tilted landing approach. For Fig. 2, the communication and sensing signal-to-noise ratios (SNRs) are fixed at 19 and 0 dB, respectively, while the required spectral efficiency is varied. For Figs. 3a and 3b, the required spectral efficiency is 5 bit/s/Hz and the sensing SNR is varied from -5 to 15 dB. Each landmark-focused beam is assigned at least $0.01P_{\max}$ so that all four coded echoes remain available for pose recovery.

Each root-mean-square error (RMSE) point is averaged over 2000 paired Monte Carlo trials. All schemes use the same array, landmark geometry, total transmit power, communication-rate requirement, observation interval, and noise realizations. In every trial, the unknown complex landmark reflectivities are eliminated rather than revealed to the estimator. The proposed design optimizes the communication and sensing powers using the pose FIM and then recovers the pose by Algorithm 1. The following benchmark schemes are considered.

- *Equal power allocation (Equal PA):* This scheme first assigns the minimum communication power required to satisfy the rate constraint and then uniformly distributes the remaining power among the four landmark-focused beams. Algorithm 1 is used for pose recovery. Comparing this scheme with the proposed power allocation therefore isolates the benefit of pose-oriented sensing-power allocation.
- *Time-division ISAC (TD-ISAC):* Following the time-division architecture in [13], communication and equal-gain landmark sensing are performed over separate portions of the observation interval. The time fraction is

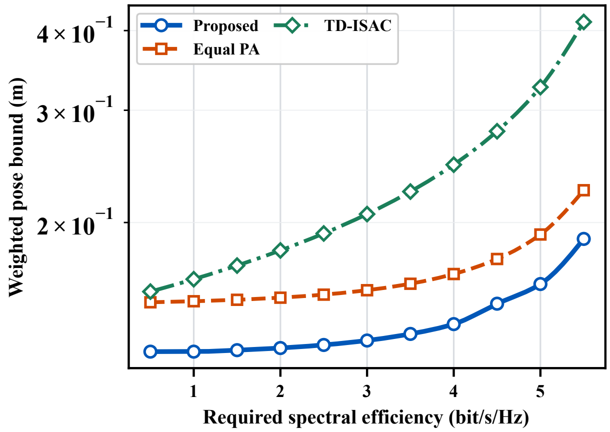


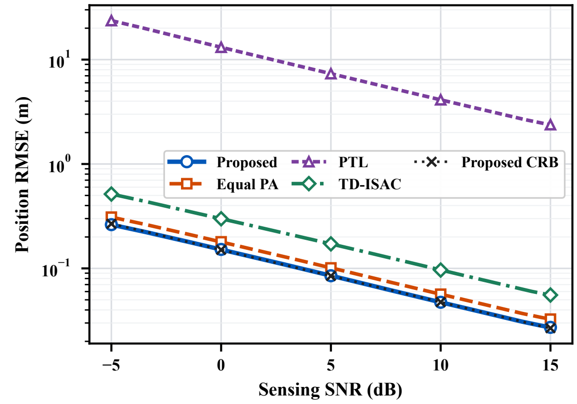
Fig. 2. Weighted pose CRB versus required spectral efficiency.

optimized under the same total resource budget, and Algorithm 1 is used for pose recovery. This benchmark quantifies the loss caused by orthogonalizing the two functions in time.
- *Point-target localization (PTL):* Motivated by conventional decoupled target localization and rigid registration [8], [11], this scheme uses exactly the communication and sensing powers obtained by the proposed power allocation, independently localizes the four landmarks, and then applies unweighted rigid registration. Hence, its comparison with the proposed pose estimator isolates the gain achieved by directly exploiting the rigid-body coupling and landmark-dependent error covariances in pose estimation.

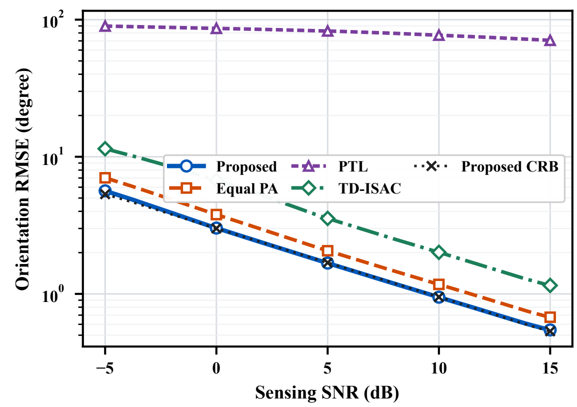
B. Pose CRB versus Communication Rate

Fig. 2 plots the weighted pose CRB versus the required spectral efficiency. It is observed that the CRBs of all schemes increase with the rate requirement. This is expected because a larger communication power is needed to satisfy a more stringent rate constraint, leaving less power for landmark sensing. The increase becomes more pronounced near the maximum feasible rate, where the available sensing-power margin is small.

It is also observed that the proposed power allocation consistently outperforms Equal PA and TD-ISAC. Equal PA ignores both the 30-dB landmark-visibility difference and the distinct rotational lever arms of the landmarks, and thus spends excessive power on echoes that provide relatively little pose information. TD-ISAC suffers an additional loss because shortening the sensing interval directly reduces the accumulated echo information. By contrast, the proposed design allocates power according to each landmark's contribution to the pose EFIM. At a required spectral efficiency of 4 bit/s/Hz, it reduces the weighted pose CRB by 16.5% and 43.8% relative to Equal PA and TD-ISAC, respectively. Since all schemes use the same total transmit power and observation duration, these gains result from more efficient resource allocation rather than additional sensing resources.



(a) Position RMSE.



(b) Orientation RMSE.

Fig. 3. Pose-estimation RMSE versus sensing SNR.

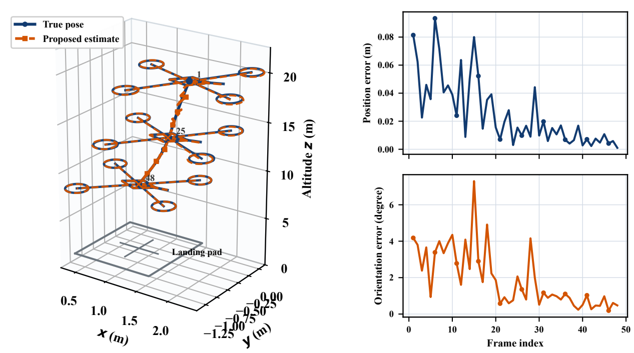


Fig. 4. Continuous UAV landing and pose-estimation errors.

C. Pose RMSE versus Sensing SNR

Figs. 3a and 3b show the position and orientation RMSEs, respectively, versus the sensing SNR. The CRB curves correspond to the proposed power allocation and are included as estimator-specific references. First, the RMSEs achieved by the proposed pose estimator remain close to their corresponding CRBs over the considered SNR range. This agreement validates both the nuisance-eliminated FIM analysis and the variable-projection pose estimator. The nearly parallel decay of the simulated RMSE and CRB curves further agrees with the expected noise-limited scaling law.

Second, the proposed joint design achieves lower position and orientation RMSEs than Equal PA and TD-ISAC. At 0-dB sensing SNR, it attains a position RMSE of 0.152 m and an orientation RMSE of 3.02°, corresponding to improvements of 15.5% and 20.3%, respectively, over Equal PA. The gain of optimized power allocation is more evident under the heterogeneous landmark visibility because Equal PA assigns the same power to landmarks with markedly different pose-information contributions. TD-ISAC performs worse because its shorter sensing interval provides fewer coherent observations.

Finally, PTL exhibits a considerably larger error despite using the same optimized transmit powers. In particular, independently localizing the weak landmark produces a highly nonuniform landmark-error profile, which is not properly handled by unweighted rigid registration. The proposed pose estimator instead estimates the six pose parameters jointly and weights the landmark echoes through their covariance information. The comparisons with Equal PA and PTL thus verify the effectiveness of the proposed transmit design and pose estimator, respectively.

D. Continuous Landing Performance

Fig. 4 illustrates one representative 48-frame UAV descent from 21 m to 7 m altitude at 0-dB sensing SNR. Each frame uses the same landmark geometry, echo-power profile, and optimized sensing-power pattern, while the estimate from the preceding frame initializes the current pose recovery. The solid blue UAVs denote the true poses, whereas the dashed orange UAVs denote the corresponding estimates at selected frames.

It is observed that the estimated trajectory closely follows the true descent in both position and orientation. Over the complete trajectory, the position and orientation RMSEs are 0.035 m and 2.48°, respectively. Relatively large frame-wise errors occur at higher altitudes, where the spherical-wave curvature across the array is weaker. As the UAV approaches the BS, the enhanced near-field curvature provides more informative range-dependent spatial signatures and the estimation errors generally decrease. This result illustrates a useful feature of near-field pose sensing that cannot be inferred from a single operating-point RMSE curve.

VI. Conclusion

This paper investigated near-field ISAC for six-dimensional pose estimation of a landing UAV equipped with calibrated coded landmarks. Based on the exact spherical-wave response and rigid-body landmark geometry, we derived the pose EFIM and CRB after eliminating the unknown landmark reflectivities, and characterized the condition for locally identifiable pose estimation. We then formulated the joint communication-and-sensing power allocation under rate and total-power constraints, transformed it into an exactly equivalent convex semidefinite program, and developed a variable-projection estimator for absolute UAV pose recovery. Numerical results demonstrated the resulting communication–pose-accuracy tradeoff and validated the proposed analytical and algorithmic designs.

Future work may extend this framework to robust power allocation and pose estimation under imperfect landmark calibration, uncertain landmark association, time-varying reflectivities, and multipath propagation. It is also of interest to exploit UAV motion over consecutive frames through trajectory-aware resource allocation and dynamic state filtering, and to validate the proposed approach using hardware prototypes and autonomous landing experiments.

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